Optical Strain based Recognition of Subtle Emotions

Sze-Teng Liong*‡, Raphael C.-W. Phan*, John See†, Yee-Hui Oh* and KokSheik Wong‡

*Faculty of Engineering, Multimedia University (MMU), Malaysia
†Faculty of Computing & Informatics, Multimedia University (MMU), Malaysia
‡Faculty of Computer Science & Information Technology, University of Malaya (UM), Malaysia
{*‡szeteng1206@hotmail.com, *raphael@mmu.edu.my, †johnsee@mmu.edu.my, *yeehui716@gmail.com, ‡koksheik@um.edu.my}

***Abstract*—This paper presents a novel method to recognize subtle emotions based on optical strain magnitude feature extraction from the temporal point of view. The common way that subtle emotions are exhibited by a person is in the form of visually observed micro-expressions, which usually occur only over a brief period of time. Optical strain allows small deformations on the face to be computed between successive frames although these subtle changes can be minute. We perform temporal sum pooling for each frame in the video to a single strain map to summarize the features over time. To reduce the dimensionality of the input space, the strain maps are then resized to a pre-defined resolution for consistency across the database. Experiments were conducted on the SMIC (Spontaneous Micro-expression) Database, which was recently established in 2013. A best three-class recognition accuracy of 53.56% is achieved, with the proposed method outperforming the baseline reported in the original work by almost 5%. This is the first known optical strain based classification of micro-expressions. The closest related work employed optical strain to spot micro-expressions, but did not investigate its potential for determining the specific type of micro-expression.**

Keywords–micro-expressions; subtle emotions; optical strain; recognition; classification; feature extraction

I. Introduction

Emotion recognition is attracting more attention in the field of psychology and computer science in recent years. In the literature, six universal emotional states have been considered: happy, fear, sad, surprise, anger and disgust. Human beings by nature inadvertently leak their emotions via facial expressions. Thus emotions are commonly analyzed via facial expressions, aimed for applications such as clinical diagnosis, national security, and interrogation. More precisely, expressions can be categorized into two main types: macro- and micro- expressions. Macro-expressions are normally voluntary expressions and occur in normal discourse. In such situations, when an emotion occurs, there is no reason for the person to hide his or her feelings. A macro-expression goes on and off the face between three fourths of a second to two seconds [1]. In contrast, a micro-epxression is a quick and involuntary facial expression which usually appears when a person unconsciously conceals a feeling [2]. Due to its short duration of occurence, which is within one twenty-fifth to one fifth of a second, micro-expressions are hard to be recognized in real time conversations [3], [4].

In 1966, Haggard et al. [5] discovered the existence of micromomentory expressions in a video recording of non-verbal communication between a therapist and a patient. Few years later, Ekman reported the finding of micro-expressions when they examined the film taken from a psychiatric patient who was trying to hide and repress her emotion. Throughout the recorded video, the patient seemed to be happy. However, when the video was thoroughly examined frame by frame, a hidden expression of despair that lasted for only two frames (one-twelved of a second) was identified. According to Ekman, micro-expressions cannot be controlled by human and they are able to reveal concealed emotions [3]. As such, detecting micro-expressions is important in community safety. A suspect interrogated by a police officer can be caught lying easily.

Ekman also implemented computerised software package, namely the Micro Expression Training Tool (METT) to automatically detect micro-expressions by focusing on the facial feature areas [6]. Later, micro-expression become a hot research topic in psychology, media and scientific research field [7], [8]. To ease the detection of micro-expressions, computers are used for automatic micro-expression detection and recognition. Although several techniques for automatic micro-expression detection and recognition have been implemented in the literature, it is still a relatively new topic and there is room for improvement.

In this paper, we propose a feature extraction method based on Black and Anandan's [9] optical flow that is robust to multiple motion to compute the optical strain magnitude for each frame in the video. Then, we apply two types of filters, which are the Wiener and Gaussian filters, to all the strain map images in order to suppress the background noise and interfering signals. Temporal sum pooling is performed for all the frames in each video to form a single strain map. This is to represent the features in a more compact and discriminative image representation. The images are then resized to a fixed and smaller dimension for standardization and to facilitate the subsequent classification process.

II. Related Work

Shreve et al. [1] used optical strain patterns as feature extractor to describe important facial motion in detecting micro-expressions. Their algorithm successfully detected the existence of micro-expressions, with one false alarm. However, two problems were pointed out in their approach: (1) the criteria set for micro-expressions (two-third of a second) is longer than most accepted duration (half a second); (2) The micro-expressions in the dataset (USF database) are posed micro-expressions rather than spontaneous ones [10]. Two years later, they tested the modified strain algorithms and experimented on two datasets (Canal-9 [11] and found videos [12]) that

contains a total of 124 spontaneous micro-expressions [13]. They achieved a 74% accuracy in spotting the rapid micro-expressions, i.e. detecting if any micro-expression exists. Note that spotting a micro-expression means that one aims to detect if any micro-expression exists. There is no determination of what specific type of micro-expression is being exhibited, which is the problem we address in this paper.

Temporal pooling is able to summarize the features over a period of time in a compact and efficient way. Boureau et al. [14] demonstrated that the performance of the recognition algorithm is closely related to the pooling step of feature extraction. In [15], Philippe et al. examined the performance of automatic annotation and ranking music radio by different combination of pooling methods(mean, maximum, minimum and variance). Pooling was also used by several researchers by vectorizing the feature descriptors to calculate the local or global bag of features [16], [17].

The Wiener filter is a classical filter which is effective in noise reduction on an image. It is a low pass filter that finds the best reconstruction of a noisy signal. In [18], it was proved that the Wiener filter is efficient in removing the noise areas by highlighting the contrast between background noise and text areas. Gatos et al. [19] implemented an adaptive Wiener method as a pre-process step of the grayscale source image based on statistics estimated from a local neighborhood of each pixel. On the other hand, the Gaussian filter is also a common filtering technique that is widely used on digital images containing facial expressions. Lien et al. [20] filters the images at the first stage before tracking the action units on the face using Facial Action Coding System(FACS) [21].

It is necessary to have a database for training purposes before developing a recognition system. However, there are limited databases which specifically record micro-expression samples. In the Polikovsky's [22] and the USD-HD [13] databases, the micro-expressions are posed ones rather than spontaneous. It is argued that posed databases may have limitations as micro-expressions are usually involuntary and hard to differentiate. In contrast, recently two databases have been presented that contain spontaneous micro-expressions: the SMIC (Spontaneous Micro-expression) database [23] and the CASME II database [24].

III. Feature Extraction

A. Optical Flow

Optical flow indicates the distribution of apparent velocities of pixels' movement between adjacent frames [25]. It computes an approximation to the motion field - the two dimension projection of the physical movement of points between two successive images [26]. It measures the spatio-temporal changes of intensity to find the matching pixel in the next frame [27]. Three assumptions are made when computing optical flow: (1) The brightness intensity of moving objects are assumed to be constant in two consecutive frames; (2) Neighboring points in the scene belong to the same surface and have similar motions; (3) Image motion of the surface patch changes gradually over time. The general optical flow equation is expressed as:

$$\nabla I \bullet \vec{p} + I_t = 0, \tag{1}$$

where *I(x,y,t)* is the image intensity function at point (x, y) at time t. ∇I = (I_x, I_y) is the spatial gradients and I_t represents the temporal gradient of the intensity function. $\vec{p}$ = $[p = d_x/d_t, q = d_y/d_t]^T$ denotes x and y components of the optical flow.

B. Optical Strain

Optical strain is able to calculate small changes on facial expressions and it expresses the relative amount of deformation of an object [28]. The two dimension displacement vector of a moving object can be expressed as $\mathbf{u}$ = $[u, v]^T$. Strain magnitude can be represented in (2) by assuming that the moving object is in small motion.

$$\varepsilon = \frac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T] \tag{2}$$

or in an expanded form:

$$\varepsilon = \begin{bmatrix} \varepsilon_{xx} = \frac{\partial u}{\partial x} & \varepsilon_{xy} = \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) \\ \varepsilon_{yx} = \frac{1}{2}(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}) & \varepsilon_{yy} = \frac{\partial v}{\partial y} \end{bmatrix} \tag{3}$$

where $(\varepsilon_{xx}, \varepsilon_{yy})$ are normal strain components and $(\varepsilon_{xy}, \varepsilon_{yx})$ are shear strain components.

In order to estimate the strain from the finite strain tensor (2), we can simplify the optical flow field (*p,q*) in (1) by approximating it to the first order derivatives as the strain components are described in displacement vector (*u,v*).

$$p = \frac{dx}{dt} = \frac{\Delta x}{\Delta t} = \frac{u}{\Delta t}, u = p\Delta t, \tag{4}$$

$$q = \frac{dy}{dt} = \frac{\Delta y}{\Delta t} = \frac{v}{\Delta t}, v = q\Delta t \tag{5}$$

where Δt is the time interval between two image frames. Since the frame rate of a video is constant, Δt is a fixed length, we can estimate partial derivative of (4) and (5) as:

$$\frac{\partial u}{\partial x} = \frac{\partial p}{\partial x}\Delta t, \frac{\partial u}{\partial y} = \frac{\partial p}{\partial y}\Delta t, \tag{6}$$

$$\frac{\partial v}{\partial x} = \frac{\partial q}{\partial x}\Delta t, \frac{\partial v}{\partial y} = \frac{\partial q}{\partial y}\Delta t \tag{7}$$

The second order derivatives can be approximated using Finite Difference Approximation.

$$\frac{\partial u}{\partial x} = \frac{u(x + \Delta x) - u(x - \Delta x)}{2\Delta x} = \frac{p(x + \Delta x) - p(x - \Delta x)}{2\Delta x} \tag{8}$$

$$\frac{\partial v}{\partial y} = \frac{v(y + \Delta y) - v(y - \Delta y)}{2\Delta y} = \frac{q(y + \Delta y) - q(y - \Delta y)}{2\Delta y} \tag{9}$$

where $(\Delta x, \Delta y)$ are preset distances of 1 pixel.

Finally, the magnitude of the optical strain can be computed as follows:

$$\varepsilon = \sqrt{{\varepsilon_{xx}}^2 + {\varepsilon_{yy}}^2 + {\varepsilon_{xy}}^2 + {\varepsilon_{yx}}^2} \tag{10}$$

IV. Proposed Algorithm

In this paper, we propose an algorithm based on the optical strain technique as the main contribution for feature extraction. After obtaining the optical strain maps, one for each pair of consecutive frames, the filtering step is then applied to all the strain map frames. The strain map information for each video are collected pixel-by-pixel using temporal pooling to form a single strain map, which serves as a temporal representative image for the respective video. The pixel intensities in each pooled strain map image is max-normalized to increase the significance of the strain information. Lastly, the normalized strain map image is resized to a pre-defined resolution to reduce the complexity and improve the computation time in the classification stage.

The effectiveness of optical strain has been shown to be better than optical flow in distiguishing expressions by comparing their magnitudes calculated over the video sequence, as described in [1]. To extract useful information from the images in the form of optical strain magnitude, vertical and horizontal motion vectors, (p, q) are first calculated along the video sequence. Optical strain magnitude, $\varepsilon_{i,j}$ is then computed over each flow field for every two consecutive frames (f_1-f_k,...,f_{k-1}-f_k, $k \epsilon$ [2, F]), where (i, j) is the pixel's coordinate in (x, y) direction and F is the number of frames. Hence each video will generate $F - 1$ strain map images.

Digital images are prone to various types of noise. The noises exist in the strain map images in regions such as background, participants' neck, clothings and wirings on headset worn. The noises can even be caused by unstable lighting conditions. They result in significant false information in the classification step. Since the micro-expressions are very fine movements on the face, the filtering process might suppress the unwanted noise together with the essential information which describe the micro-expressions. Therefore, the parameters of the filters need to be adjusted for optimal performance.

Temporal pooling is then performed by summing the magnitude of optical strain maps for each pixel position in the map, across all the maps from the starting frame to the last F^{th} one. The purpose of pooling is to reduce the number of features to a more compact image representation in a lower dimension. For each video, the pixel intensities of the pooled strain image, $s_{i,j}$ are divided by their respective number of maps, since all the videos have different number F of frames and therefore number of maps:

$$s_{i,j} = \frac{1}{F-1} \sum_{f=1}^{F-1} \varepsilon_{i,j}, i = 1 \ldots X, j = 1 \ldots Y \tag{11}$$

where X and Y denote the number of rows and columns for each frame/map (along X-Y plane).

The pooled strain images are maximum-normalized to improve the range of the pixel intensity values as each pixel will be treated as a feature. Although the dataset consists of cropped faces of the subjects, the resolution of the frames is considered high in the feature descriptor point of view. Therefore, all the pooled strain maps were resized to 50×50 pixels by bilinear interpolation to reduce the feature length and computation time in the classifier.

The whole process of the proposed algorithm is illustrated in Figure 1.

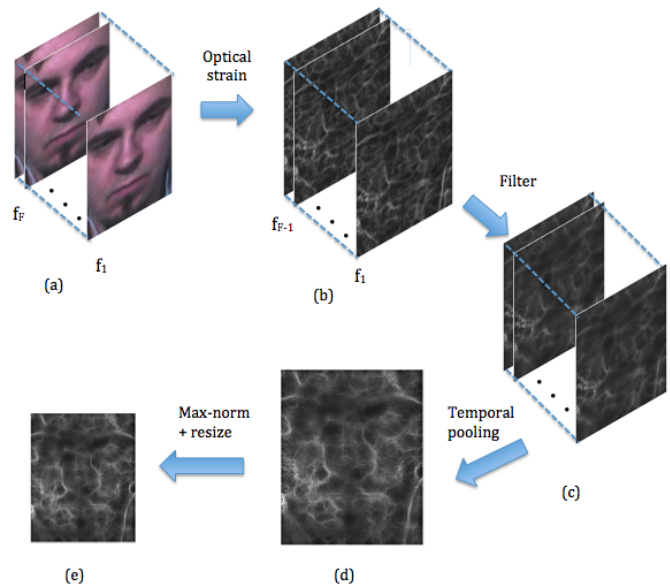

Fig. 1. Feature extraction for a video sequence: (a) Original images (b) Optical strain map images (c) Images after passing through filter (d) Pooled strain map image (e) After applying maximum-normalization and resize on the pooled strain image

V. Experiment

A. Dataset

The dataset used for experimentation is the SMIC (Spontaneous Micro-expression) database [23]. The database consists of 16 subjects (mean age of 28.1 years, 6 females and 10 males) with a total of 164 video clips. The video clips are classified into three main categories: surprise, positive (happy) and negative (sad, fear, disgust). Each video clip only contains a type of micro-expression. The videos were recorded using a high speed camera (PixeLINK PL-B774U) with a resolution of 640×480 and frame rate of $100 fps$. The candidates were asked to watch a series of emotional movie films from the computer with the camera placing on top of the monitor. The candidates would try their best to hold their genuine feelings by showing poker faces along the video clips. This is because micro-expression always reveal the real emotion of the person. If the researchers guess the feeling of the candidates correctly, the candidates would have to fill in a questionnaire which have more than 500 boring questions. The recorded clips were analyzed and selected by two annotators by following the advice of Ekman [3] (view the video frame-by-frame, then with increasing speed) and compare with the participants self-reported emotion. The dataset used 0.5 seconds as the threshold

TABLE I. COMPARISON OF THE LEAVE-ONE-SUBJECT-OUT RECOGNITION RESULTS BETWEEN THE BASELINE AND PROPOSED OPTICAL STRAIN METHOD ON THE SMIC DATABASE.

	SVM kernel	
Methods	**RBF (%)**	**Linear (%)**
Baseline (TIM10 + LBP-TOP ($8 \times 8 \times 1$))	48.78	48.78
OS + Gaussian filter	46.10	47.36
OS + Wiener filter	**51.24**	**53.56**

Fig. 2. Example of a surprise expression in the SMIC database

for micro-expressions. The best performance obtained was 48.78% by using leave-one-subject-out cross-validation SVM classifier. All the videos were first interpolated using temporal interpolation model (TIM) [29] to speed up the feature extraction (LBP-TOP) process. Figure 2 shows a sequence of frames of a surprise expression in the database.

B. Results and Discussion

Experiments were carried out on the SMIC database to perform a three-class micro-expression recognition (Positive vs. Negative vs. Surprise). To evaluate the performance of our implementation, we adopt the baseline result from the original SMIC paper [23], which employed a combination of techniques including TIM, LBP-TOP and SVM. The best classification performance was achieved using TIM to down-sample the frames of each video to 10 key frames. The features were extracted using a block-based LBP-TOP that consists of $x \times y \times t$ blocks (denoting the row, column and temporal dimensions respectively) to segment the frames. The LBP-TOP radii of the (XY, YT, XT) planes used in the original work are $(1, 1, 3)$ respectively. However, the authors did not mention which kernel they used in the SVM classification. Therefore, we assumed that the best result for both RBF and linear kernels is 48.78%.

In the filtering step, we used two types of low-pass filters — the Gaussian filter and the Wiener filter, to reduce the background noise and irrelevant signals. The comparison of the recognition results using both types of filters can be observed in Table I. We note that different combination of parameters for the Gaussian filter produces a different effect on the strain image. By experiments, the parameters that produced the best result are $\sigma = 1.2$ and filter size = 5×5 pixels, with an accuracy rate of 47.36% (Linear kernel), which is slightly lower than that of the baseline. However, when the Wiener filter is used, the accuracy of recognition increases to 53.56% (linear kernel), which is an improvement of 4.78% compared to the baseline performance. Likewise, its performance on the RBF kernel (for SVM) also surpasses that of the baseline. Empirically, we found that the filter size that generates the best performance on the Wiener filter is 10×10 pixels.

The fact that Wiener filter is able to outperform Gaussian filter is because it is an optimum adaptive filter that is based on statistical approach. It can tailor itself to different local image variance. For example, when the variance at the particular location is large, the filter will carry out less smoothing and vice versa. It operates better than Gaussian filter in images when the noise model resembles that of Gaussian white noise or salt and pepper noise [30]. Meanwhile, the Gaussian filter can effectively remove noise drawn from a normal distribution and control the amount of blurring effect by adjusting the standard deviation, σ. Figure 3 illustrates the comparison of the strain images before and after applying Gaussian and Wiener filter. It can be seen that facial details are much more reduced in terms of the optical strain magnitudes and it blurs the image when using Gaussian filter compared to Wiener filter. This explains why Wiener filter performs better than Gaussian filter in optical strain image, as the essential information are preserved in the strain image.

Fig. 3. Sample strain image: (left) Original image; (middle) After applying Gaussian filter; (right) After applying Wiener filter. Enlarge this figure to observe the loss of information on the Gaussian filtered strain image.

VI. CONCLUSION

In this paper, we have presented a novel method for the automatic recognition of facial micro-expressions on the SMIC database. The proposed method, which employs optical strain magnitudes for feature extraction, including a combination of other techniques such as filtering, temporal pooling, maximum normalization, and re-sizing, has achieved a more promising result compared to the reported baseline result. The optical strain magnitudes used can robustly compute the temporal motion details for each frame. Hence, our method is able to recognize the micro-expressions of the participants independently due to its ability to capture subtle or rapid motions on the face. For future research, the optical strain feature extractor can be used in various applications such as clinical diagnosis and national security to reveal the presence of subtle emotions and classify them thereafter. Furthermore, the type of filter and its parameters can be altered to optimize its effect on the algorithm.

ACKNOWLEDGMENT

This work is supported in part by Telekom Malaysia under the project UbeAware and by University Malaya Research Collaboration Grant (Title: Realistic Video-Based Assistive Living, Grant Number CG009-2013) under the purview of the University of Malaya Research.

REFERENCES

[1] Shreve, M., Godavarthy, S., Manohar, V., Goldgof, D., Sarkar, S.: Towards macro-and micro-expression spotting in video using strain patterns. In: Applications of Computer Vision (WACV). (2009) 1–6

[2] Ekman, P., Friesen, W.V.: Nonverbal leakage and clues to deception. Journal for the Study of Interpersonal Processes **32** (1969) 88–106

[3] Ekman, P.: Lie catching and microexpressions. The philosophy of deception (2009) 118–133

[4] Porter, S., ten Brinke, L.: Reading between the lies identifying concealed and falsified emotions in universal facial expressions. Psychological Science **19.5** (2008) 508–514

[5] Haggard, E.A., Isaacs, K.S.: Micromomentary facial expressions as indicators of ego mechanisms in psychotherapy. In Methods of research in psychotherapy (1966) 154–165

[6] Ekman, P.: Micro-expression training tool (METT). (2002)

[7] Gottman, J.M., Levenson, R.W.: A twofactor model for predicting when a couple will divorce: Exploratory analyses using 14year longitudinal data. Family process **41.1** (2002) 83–96

[8] Warren, G., Schertler, E., Bull, P.: Detecting deception from emotional and unemotional cues. Journal of Nonverbal Behavior **33.1** (2009) 59–59

[9] Black, M.J., Anandan, P.: The robust estimation of multiple motions: Parametric and piecewise-smooth flow fields. Computer vision and image understanding **63.1** (1996) 75–104

[10] Yan, W.J., Wang, S.J., Liu, Y.J., Wu, Q., Fu, X.: For micro-expression recognition: Database and suggestions. Neurocomputing **136** (2014) 82–87

[11] Vinciarelli, A., Dielmann, A., Favre, S., Salamin, H.: Canal9: A database of political debates for analysis of social interactions. In: In Affective Computing and Intelligent Interaction and Workshops. (2009) 1–4

[12] Ekman, P. In: Telling Lies: Clues to Deceit in the Marketplace, Politics, and Marriage. W. W. Norton and Company (2009) WW Norton and Company.

[13] Shreve, M., Godavarthy, S., Goldgof, D., Sarkar, S.: Macro-and micro-expression spotting in long videos using spatio- temporal strain. In: Automatic Face, Gesture Recognition and Workshops. (2011) 51–56

[14] Boureau, Y.L., Ponce, J., LeCun, Y.: A theoretical analysis of feature pooling in visual recognition. In: In Proceedings of the 27th International Conference on Machine Learning. (2010) 111–118

[15] Hamel, P., Lemieux, S., Bengio, Y., Eck, D.: Temporal pooling and multiscale learning for automatic annotation and ranking of music audio. In: International Society for Music Information Retrieval Conference. (2011) 729–734

[16] Zhang, J., Marszaek, M., Lazebnik, S., Schmid, C.: Local features and kernels for classification of texture and object categories: A comprehensive study. International journal of computer vision **73.2** (2007) 82–87

[17] Sivic, J., Zisserman, A.: Video google: A text retrieval approach to object matching in videos. In: In Computer Vision, Ninth IEEE International Conference. (2003) 1470–1477

[18] Jain, A.K. In: Fundamentals of digital image processing. Prentice Hall; 1 edition (1989) Prentice-Hall.

[19] Gatos, B., Pratikakis, I., Perantonis, S.J.: An adaptive binarization technique for low quality historical documents. In: Document Analysis Systems VI. (2004) 102–113

[20] Lien, J.J.J., Kanade, T., Cohn, J.F., Li, C.C.: Detection, tracking, and classification of action units in facial expression. Robotics and Autonomous Systems **31.3** (2000) 131–146

[21] Ekman, P., Friesen, W.V.: Facial action coding system. Consulting Psychologists Press (1978)

[22] Polikovsky, S., Kameda, Y., Ohta, Y.: Facial micro-expressions recognition using high speed camera and 3d-gradient descriptor. In: Crime Detection and Prevention. (2009) 16–16

[23] Li, X., Pfister, T., Huang, X., Zhao, G., Pietikainen, M.: A spontaneous micro-expression database: Inducement, collection and baseline. In: Automatic Face and Gesture Recognition. (2013) 1–6

[24] Yan, W.J., Wang, S.J., Zhao, G., Li, X., Liu, Y.J., Chen, Y.H., Fu, X.: CASME II: An improved spontaneous micro-expression database and the baseline evaluation. PLoS ONE **9** (2014) e86041

[25] Horn, B.K., Schunck, B.G.: Determining optical flow. In: International Society for Optics and Photonics. (1981) 319–331

[26] Barron, J.L., Fleet, D.J., Beauchemin, S.S.: Performance of optical flow techniques. International journal of computer vision **12.1** (1994) 43–77

[27] Jain, R., Kasturi, R., Schunck, B.G.: Machine vision. Volume 5. McGraw-Hill Education (1995)

[28] Godavarthy, S.: Microexpression spotting in video using optical strain. Masters thesis, University of South Florida (2010)

[29] Zhou, Z., Zhao, G., Guo, Y., Pietikainen, M.: An image-based visual speech animation system. Circuits and Systems for Video Technology **22.10** (2012) 1420–1432

[30] Srivastava, C., Mishra, S.K., Asthana, P., Mishra, G.R., Singh, O.P.: Performance comparison of various filters and wavelet transform for image de-noising. IOSR Journal of Computer Engineering **10.1** (2013) 55–63